

Pack — Build Summary

Multi-agent AI hunt orchestrator · the wolf-pack that researches, drafts, and digs things up for you.

Branch `tobiloba/engine-spine` · Pull Request #5 · ~75 commits · 20 June 2026

This branch stands up the **real backend engine**, a **frontend wired live to it**, and a **conversational front door** where Alpha (the pack leader) talks to you like a normal assistant and only sends the pack on a real job. It runs end-to-end against the real Qwen model.

The front door — Alpha talks first

The headline change. Instead of turning every keystroke into a "hunt," the entry is a real chat:

- **Alpha converses like ChatGPT** — answers questions, explains himself, handles small talk, even simple facts. He only proposes a hunt when there's an actual task (research, draft, review, compare, summarize, dig something up).
- **Clarify → confirm → run.** When he understands the task he proposes it and waits — nothing launches until you click *Send the pack*. Then the conversation carries straight into the hunt.
- This shape is backed by deep research into how leading agents (OpenAI/Perplexity "deep research", etc.) handle intake — clarify only when it matters, confirm once, then go.

Backend — the engine

- **Transactional outbox** (not dual-write): Postgres is the single source of truth; a relay ships committed events to Redis Streams — at-least-once, idempotent at the reducer. No lost or double-counted events.
- **Per-hunt supervisor** with a dual ledger (Task + Progress) coordinating the wolves; a command queue and the 202-accepted command contract.
- **Qwen client** via the OpenAI SDK at the DashScope endpoint (tiers max / plus / flash), with an offline fallback so it still boots with no API key.
- **17 REST endpoints** — the full command + read surface, including the new conversational `/hunts/intake` and `/hunts/<id>/ask`.
- **26 automated tests** passing (outbox relay, event sequencing, offline hunt, pricing, repository, contract).

Frontend — every screen, live

Area	What it does
The Door	Real chat with Alpha — typed reveal, his avatar, confirm-before-launch, full scrollable history.
Hunt screen	State-driven: review & <i>Send the pack</i> → live canvas → the Return (deliverable takes center).

Territory canvas	The pack as circular nodes with a 7-state look; edges show handoffs; idle pack shown before approval.
Plan rail	Carries the conversation, spend meter, Boundary + Send, the one Hold card, Stop, Ask Alpha.
Return	The finished brief — Download, Copy, Save as Instinct, links to Tracks & Scorecard.
Tracks / Scorecard	The full event log timeline, and Lone Wolf vs Pack comparison.
The Den	Past hunts + saved Instincts, pulled from the engine.

A pure event-stream reducer drives the store; the UI never computes truth, it renders it. Live updates arrive over a WebSocket from the gateway.

Developer experience

- **Postman collection** for all 17 endpoints — folders, example bodies, auto-captured `huntId` , environment, README, and a generator that keeps it in sync (`docs/postman/`).
- **Local setup guide** + Docker Compose for Postgres & Redis (see the companion "Run It Locally" doc).

By the numbers

Backend endpoints	17	Backend tests	26 passing
Commits	~75	Files changed	66 (+5,286 / -526)
Frontend	build + lint green	Live-verified	intake → hunt → returned

Honest next steps (not yet built)

Real Lone-Wolf-vs-Pack benchmark · click-to-trace provenance · resume-from-checkpoint · real web search for the Scouts · file uploads. None of these are faked in the UI.